



Geomagnetic Disturbance Expected Today into Saturday

This briefing is tailored for reliability coordinators, balancing authorities, transmission operators, ISOs/RTOs, and operational coordination groups.

A train of CMEs is expected to reach Earth today into Saturday. Significant geomagnetic activity is expected, and warning-level conditions may persist through the weekend. Current analysis continues to show an 80-90% chance that two or more CMEs will impact Earth. G3 to G4 geomagnetic conditions are expected, with a 50-60% chance of G5 conditions.

STORM-LEVEL PROBABILITY

PROBABILITY OF G3

Strong geomagnetic storming

- G3 conditions are expected during the event window.
- Heightened monitoring is warranted for susceptible systems.
- Regional response may vary with local ground conductivity and system posture.

80-90%

PROBABILITY OF G4

Severe geomagnetic storming

- Severe intervals are possible if CME coupling is favorable.
- Voltage alarms, voltage control issues, or GIC response may become more likely.
- Sustained storming would increase operational concern.

50-60%

PROBABILITY OF G5

Extreme geomagnetic storming

- Extreme conditions remain possible but less likely.
- Concern increases if multiple CME structures arrive close together.
- Use observed SWPC alerts and local indicators to reassess readiness.

20-30%

KEY TIMING

Timing Window	Forecast Message	Operational Meaning
Fri afternoon-Fri evening UTC	60-70% chance of initial CME arrival	Begin heightened monitoring if not already active
Fri night-early Sat UTC	20-30% chance of later arrival	Maintain readiness through the overnight period
Sat into Sun	Additional CME influences likely	Prepare for prolonged or repeated disturbance periods
After peak conditions	Possible de-escalation to Statement	Continue monitoring until conditions clearly decrease

DURATION CONCERN

With multiple CMEs expected to arrive, elevated geomagnetic activity could persist for 24 to 48 hours. There is a 70-80% chance of disturbed conditions lasting 24 hours or longer, and a 50-60% chance of disturbed conditions lasting 48 hours or longer.

CONFIDENCE AND UNCERTAINTY

Confidence is medium to high that significant geomagnetic activity will occur. Confidence is lower in exact peak intensity and duration. The most important uncertainty is how strongly Earth's magnetic environment responds after CME arrival and whether additional CME structures prolong the event.

POTENTIAL GRID-RELEVANT IMPACTS

- Voltage alarms or voltage control issues.
- Increased geomagnetically induced current potential.
- Weak to moderate power system fluctuations.
- Greater concern if severe storming is sustained over multiple hours.
- Possible repeated periods of enhanced geomagnetic activity as additional CME structures arrive.

WHAT OPERATORS SHOULD WATCH

- SWPC Warning Updates.
- Observed Space Weather Alerts, especially G4 or G5 conditions.
- Solar wind conditions after CME arrival.
- Regional geoelectric field guidance, if available.
- Internal GIC monitors, transformer response, voltage alarms, and local system indicators.
- Any reports from neighboring systems or reliability coordination channels.

ACTIVE PRODUCTS

- Space Weather Warning: In effect.
- Observed Space Weather Alerts: Expected if thresholds are reached.
- Partner briefings and Warning Updates: Continuing through the event.

NEXT UPDATE

SWPC will continue issuing Warning Updates at least every 12 hours while the Warning remains in effect, or sooner if observations, impacts, or forecast guidance change.

For additional information, monitor spaceweather.gov and SWPC decision-support guidance.